

DIABETIC KETOACIDOSIS

The background of the slide is a dark blue gradient. On the right side, there is a series of overlapping, three-dimensional rectangular blocks. One block is a light blue color, and another block below it is an orange color. The rest of the blocks are in various shades of dark blue and black, creating a sense of depth and geometric complexity.

Introduction of DKA

- Diabetic ketoacidosis (DKA) is a potentially life-threatening complication in people with diabetes mellitus. It happens predominantly in those with type 1 diabetes, but it can occur in those with type 2 diabetes under certain circumstances.
- DKA results from a shortage of insulin; in response the body switches to burning fatty acids and producing acidic ketone bodies that cause most of the symptoms and complications

Definition of DKA

- DKA can be defined as a blood glucose level usually $>250\text{mg/dl}$, $\text{pH} < 7.25$ & plasma bicarbonate level of 15mEq/l or less.
- Severe DKA is define as a pH of 7.1 or less & bicarbonate level of 10mEq/l or less.
- DKA is a common & potentially life threatening, acute complication of IDDM. Mortality rate may be as high as 6-10%.
- Hyperglycemia: $\text{BG} > 11\text{ mmol/l}$
- Hyperketonemia or hyperketonuria
- Metabolic acidosis, venous blood gas : $\text{PH} < 7.3$ +/or Bicarbonate $< 15\text{ mmol/l}$

Cause of DKA

- DKA is brought on by a lack of insulin, the hormone generated by your pancreas to help your body produce glucose.
- Without insulin, your body begins to break down fat in attempt to get the energy it needs, a process that ultimately results in a dangerous buildup of acids.
- When this happens, your body can go into shock, and the acid buildup may causes swelling in the brain. It can induce a coma and become life-threatening.

Severity of DKA

Degree of acidosis

- Moderate
- Severe

PH

<7.2

<7.1

Bicarbonate

<10mmol/l

<5mmol/l



Precipitating Factors

- Missed or inadequate insulin doses
- Intercurrent illness; infection or stress
- Surgery or trauma

Symptoms and Signs

- Polyuria
- Polydipsia
- Nocturnal enuresis
- Dehydration
- Rapid deep breathing
- Abdominal pain
- Acetone smell of breath
- Disturbed level of consciousness

Investigation

- Serum ~ Urine
- Random blood glucose ~glucose
- Urea and Electrolytes ~ketone
- Bone profile
- CBC
- Osmolality
- Culture

Monitoring

Clinical

- Keep NPO, with 24 hours intake/output chart
- Cardiac monitoring
- Hourly heart rate (HR), respiratory rate (RR), blood pressure (BP), level of consciousness
- Hourly neurological assessment for warning signs and symptoms of cerebral edema
- 4 Hourly temperature checking
- Nasogastric tube (NGT) for unconscious or vomiting patient

Biochemical

- Hourly BG by reflow check
- Urea and electrolytes on presentation, then after 2 hours then 4-6 Hourly as indicated till patient is stable
- 2 Hourly venous blood gas
- Urine, whenever passed, to be checked for ketones

Management

Emergency Assessment

- ABC
- Clinical evaluation to confirm the diagnosis and carefully look for source of infection
- Assess clinical severity of dehydration. It should be 5-10% for usual DKA presentation. However, if > 10% dehydration is suggested, it should be justified by the presence of weak or impalpable peripheral pulse, hypotension or oliguria.
- In rare case of shock, the circulatory volume should rapidly be restored with isotonic saline in 20ml per kg bolus.
- Assess level of consciousness.

Indications for PICU Admission

- Young age ≤ 2 years
- Severe presentation
- Disturb level of consciousness at presentation or during management.
- Severe electrolytes abnormalities (e.g.: hyponatremia)
- Suspected cerebral edema at presentation or during management

Fluid Therapy

- Based on clinical assessment of 5-10% dehydration, start with 10ml/kg of normal saline over 1-2 hours and may be repeated if necessary.
- Subsequent fluid management (maintenance over 24 hours plus deficit over 36-48 hours) should be with an isotonic solution (0.9% saline) for at least 4-6 hours.
- After 6 hours fluid can be changed to half normal saline (0.45% NaCl).
- The total fluid per day should not exceed 1.5-2 times the usual daily maintenance requirement based on weight
- The patient should be assessed frequently for the correction of dehydration and the fluid is adjusted
- Monitor BG hourly, when it falls < 15mmol/l change the fluid to 0.45%NS + 5% Dextrose

Insulin Therapy

- Regular insulin IV infusion to be started 1 hour after starting fluid replacement therapy, i.e. after the patient has received initial volume expansion.
- Infusion rate: 0.05-0.1 unit/kg/h (e.g., one method is to dilute 50 units regular (soluble) insulin in 50ml. normal saline, 1 unit-1mL).
- Accept drop of glucose up to 8mmol/l in the first hour and no more than 5-6mmol/l thereafter.
- If BG drops fast and acidosis persist, change the IV Dextrose concentration to 10% or if needed 19% depend on blood glucose.
- If acidosis is corrected and BG > 15mmol/l persist, reduce the rate of insulin to 0.05u/kg/hr.
- When the acidosis is corrected and patient tolerates oral feed, subcutaneous (SC) Insulin R started. SC insulin is started 30 minutes before a meal and IV insulin stopped 30 minutes a insulin.

Correction of electrolyte disturbance

- Potassium (K) supplement of 40mmol/L should be added to the IVF if there is no concerns about renal failure and patient passed urine in the preceding 4 hours.
- Adjust K when serum electrolytes available:
 - i. If K dropped $< 3.5\text{mmol/l}$, increase supplement to 60mmol/l.
 - ii. If K $> 5\text{mmol/l}$, reduce supplement to 20mmol/l
- K supplement may be given as 50% KCL and 50% KPO₄ in case of severe hypophosphatemia. Monitor serum calcium and Phosphate in such cases.

Note

- Hyperglycemia cause a dilutional effect, depressing serum sodium concentration:
 - Corrected sodium $\text{mmol/l} + \frac{\text{Glucose (mmol/l)} - 5.6}{5.6} \times 1.6$
- 48 hours.
- If corrected sodium $> 150 \text{mmol/l}$, hypernatremia state exist and fluid therapy should be given over

Correction of acid base Imbalance

- Acidosis gets corrected spontaneously with the correction of dehydration. Routine bicarbonate (HCO_3) administration in case of DKA is not indicated and can lead to complications.
- Indication of HCO_3 if:
 - $\text{PH} < 6.9$ after initial resuscitation
 - Patient having shock
- Life-threatening hyperkalemia and arrhythmias
- Dose of bicarbonate: 1-2mmol/kg over 2 hours.

Management of Cerebral edema in DKA

Background

- Cerebral edema in diabetic ketoacidosis (CEDKA) is a serious complication and the incidence of clinically overt cerebral edema is 0.5-0.9% and the mortality rate is 21-24%. Mental status abnormalities (GCS scores < 14), however, occur in approximately 15% of children treated for DKA and are associated with evidence of cerebral edema on neuroimaging.

Risk Factors

- Younger age
- New onset diabetes
- Longer Duration of symptoms

Potential risk Factors at diagnosis or during treatment of DKA

- Greater hypocapnia at presentation after adjusting for degree of acidosis
- Increased serum urea nitrogen at presentation
- More severe acidosis at presentation
- Bicarbonate treatment for correction of acidosis
- An attenuated rise in serum sodium concentration
- Greater volumes of fluid given in the first 4h
- Administration of insulin in the first hour of fluid treatment

Signs and symptoms of Cerebral edema include

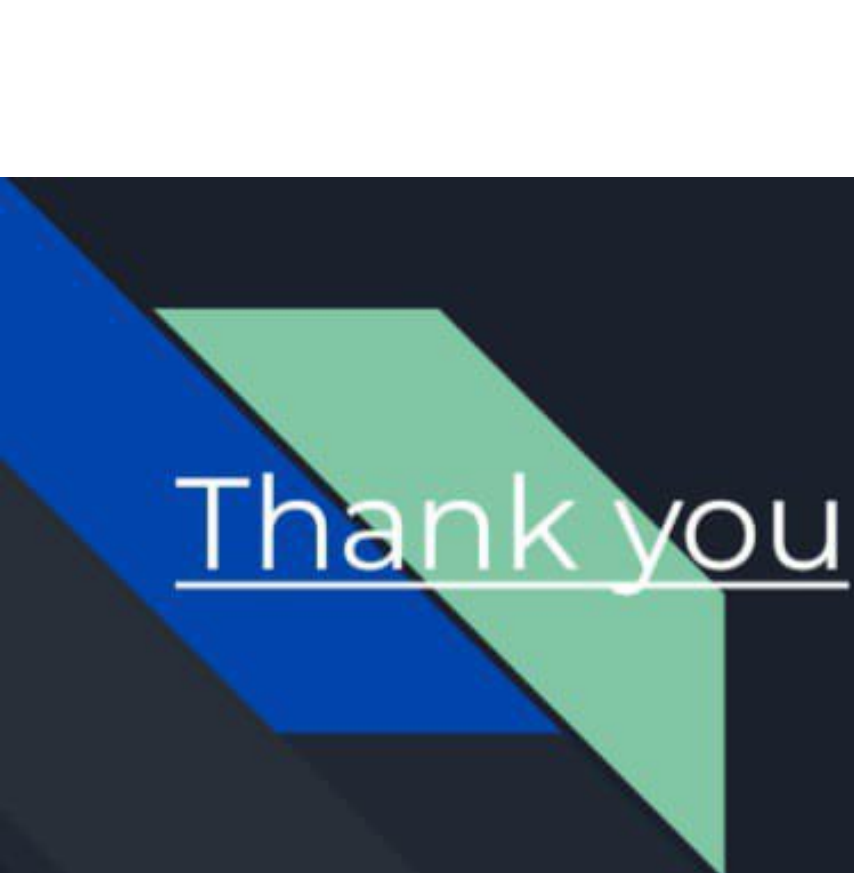
- Headache and slowing of heart rate
- Change in neurological status (restlessness, irritability, increased drowsiness, and incontinence)
- Specific neurological signs (e.g., cranial nerve palsies, papilledema)
- Rising blood pressure
- Decreased O₂ saturation

Treatment

- • Reduce the rate of fluid administration by one-third.
- Give mannitol, 0.5-1g/kg IV over 10-15min, and repeat if there is no initial response in 30min to 2h
- Hypertonic saline (3%), suggested dose 2.5-5mL/kg over 10-15min, may be used as an alternative to mannitol, especially if there is no initial response to mannitol.
- Hyperosmolar agents should be readily available at the bedside.
- Elevate the head of the bed to 30°.
- Intubation may be necessary for the patient with impending respiratory failure.
- After treatment for cerebral edema has been started, cranial imaging may be considered as with any critically ill patient with encephalopathy or acute focal neurologic deficit. The primary concern is whether the patient has a lesion requiring emergency neurosurgery (e.g., intracranial hemorrhage) or a lesion that may necessitate anticoagulation (e.g., cerebrovascular thrombosis).

References

Title of book/journal/ articles/website.	Author	Year of Publication	Page
ISPAD clinical practice Consensus Guidelines		2014	
Pediatric Diabetes.	Wolfsdorf JI	2014	154-174

A decorative graphic on the left side of the slide. It consists of a dark blue background with a diagonal band of blue and green geometric shapes. The blue shape is a parallelogram, and the green shape is a trapezoid, both overlapping each other and the background.

Thank you

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